

IN4MATX 285:

Interactive Technology Studio

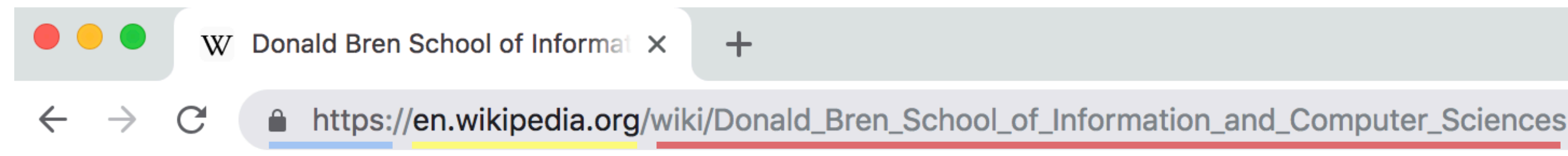
**Practice: Frontend vs. Backend
Development**

Today's goals

By the end of today, you should be able to...

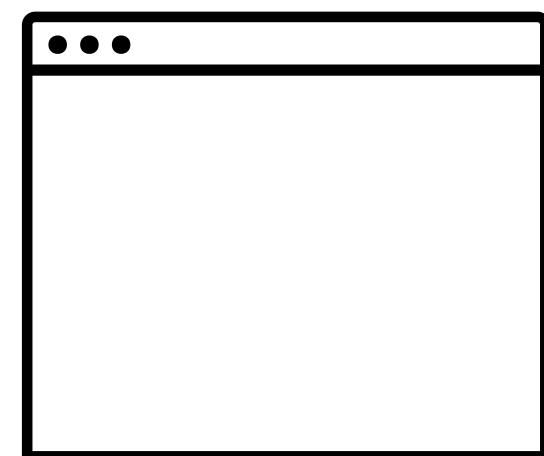
- Define frontend and backend, as well as client and server
- Differentiate the roles of browsers and web servers in presenting web content
- Articulate the more appropriate target for a particular development task

Using the internet



Protocol	Host	Resource
(how to handle info)	(who has info)	(what info you want)

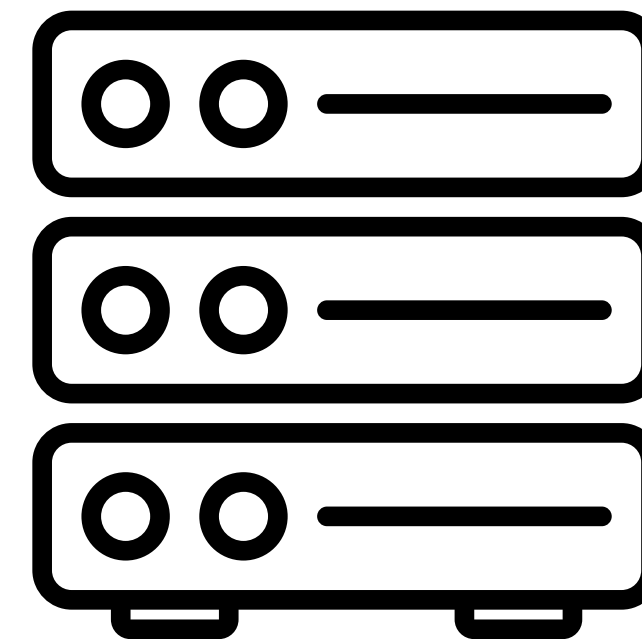
“Hey Wikipedia, I’d like to see the page for the school of ICS!”



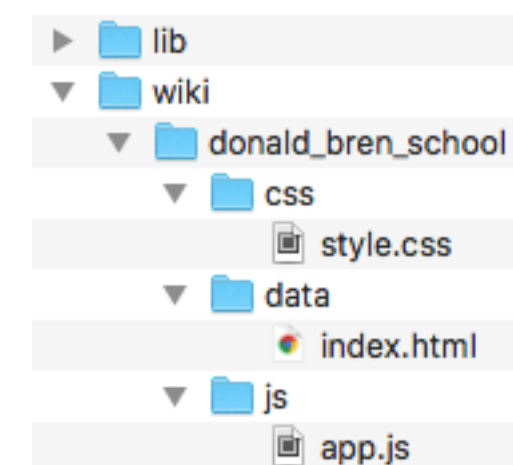
Your browser

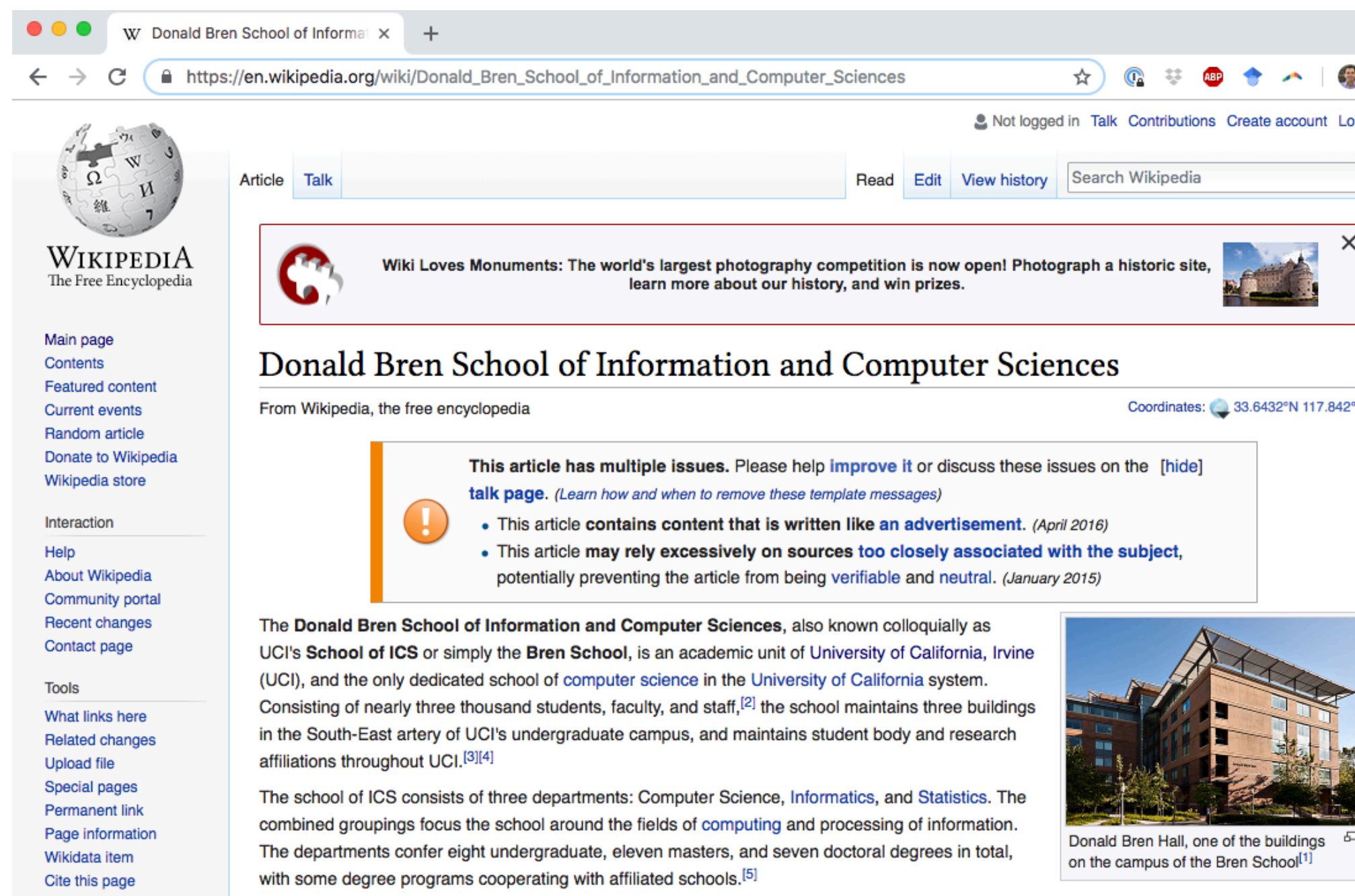
Request

Response



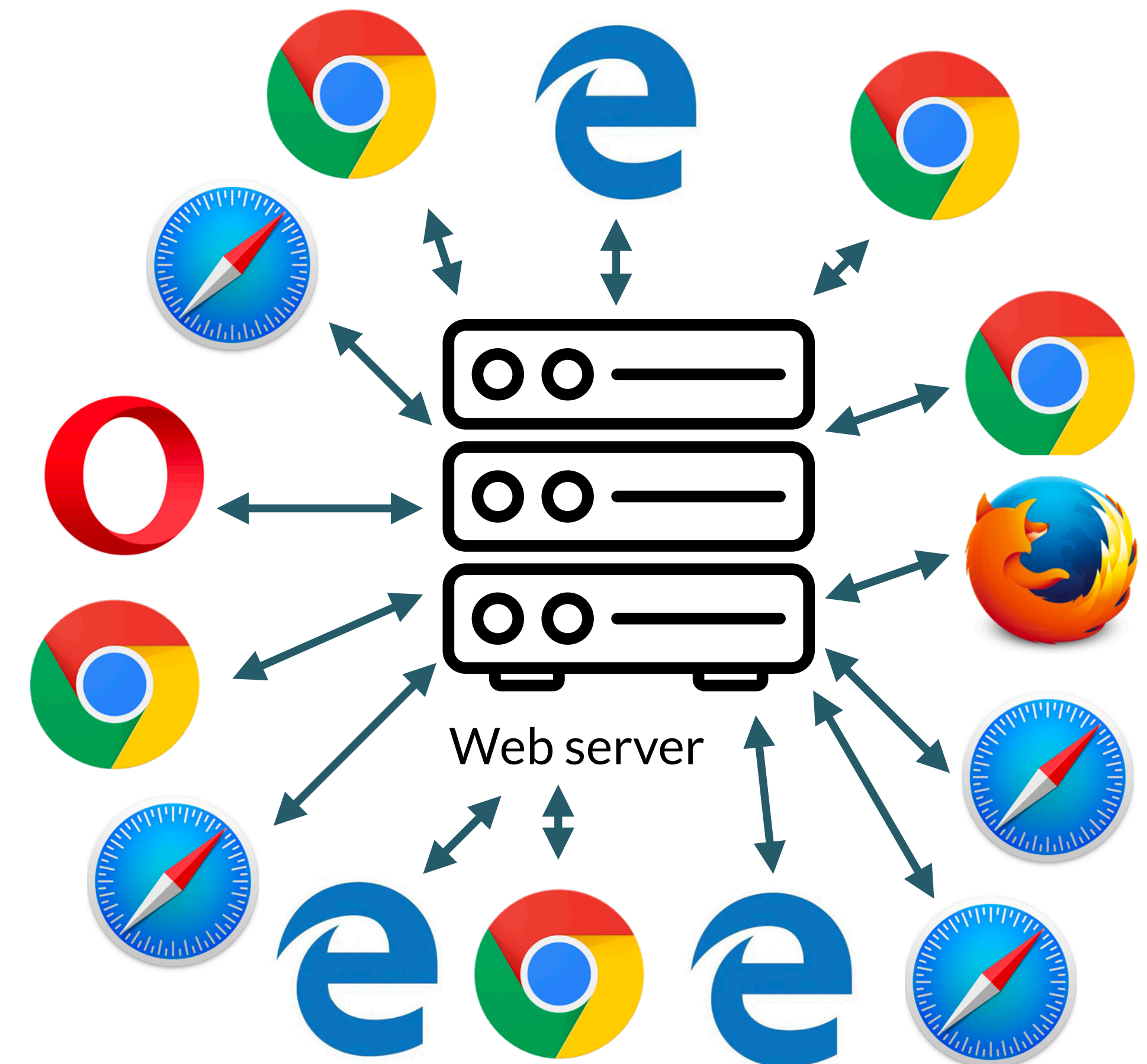
Web server





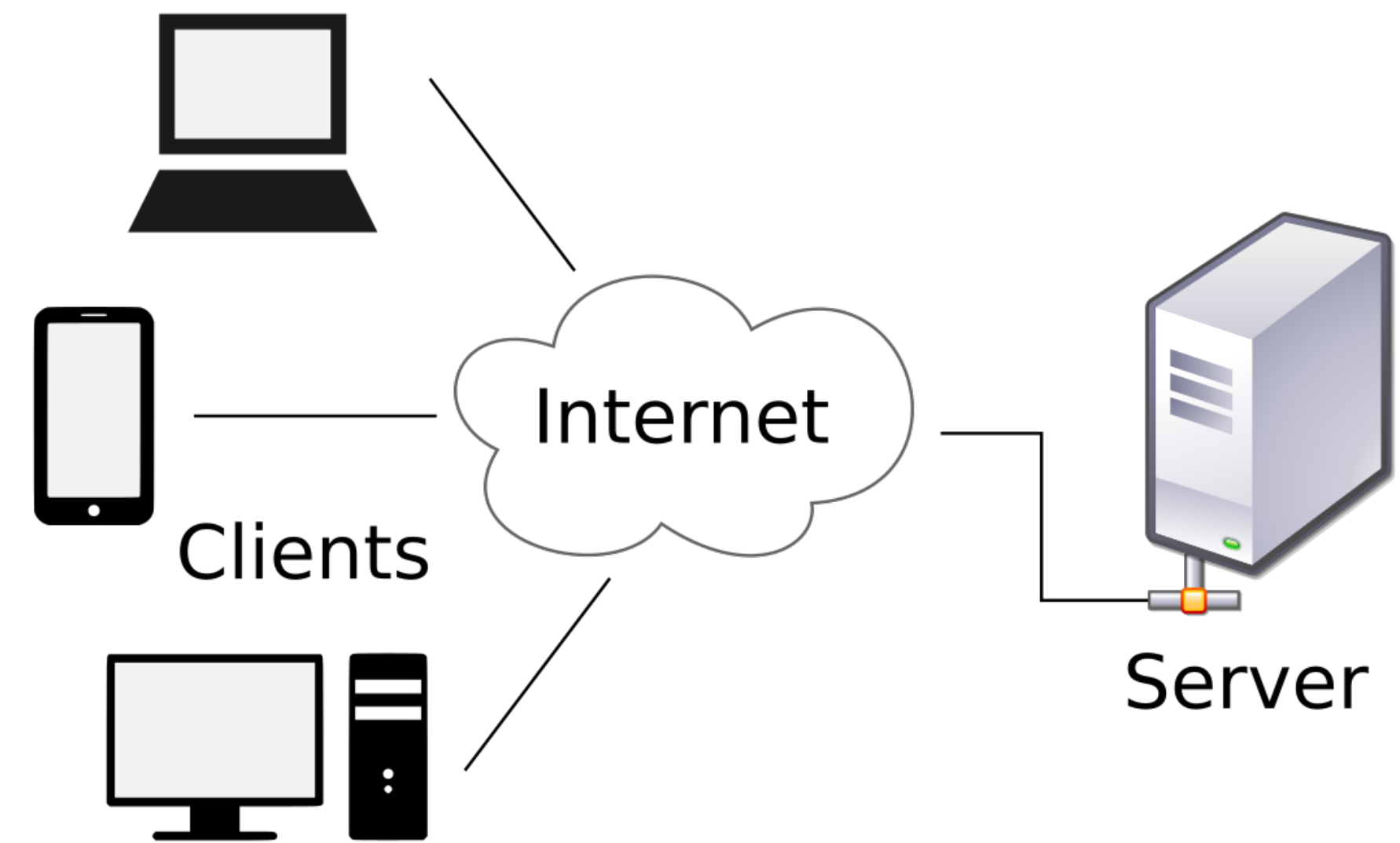
Using the internet

- Somewhere there exists a *webserver* that is sending you the files
- Your browser is the *client*
- Lots of clients, few webserver
 - Wikipedia gets 4,000 page views *every second*



Clients versus Servers

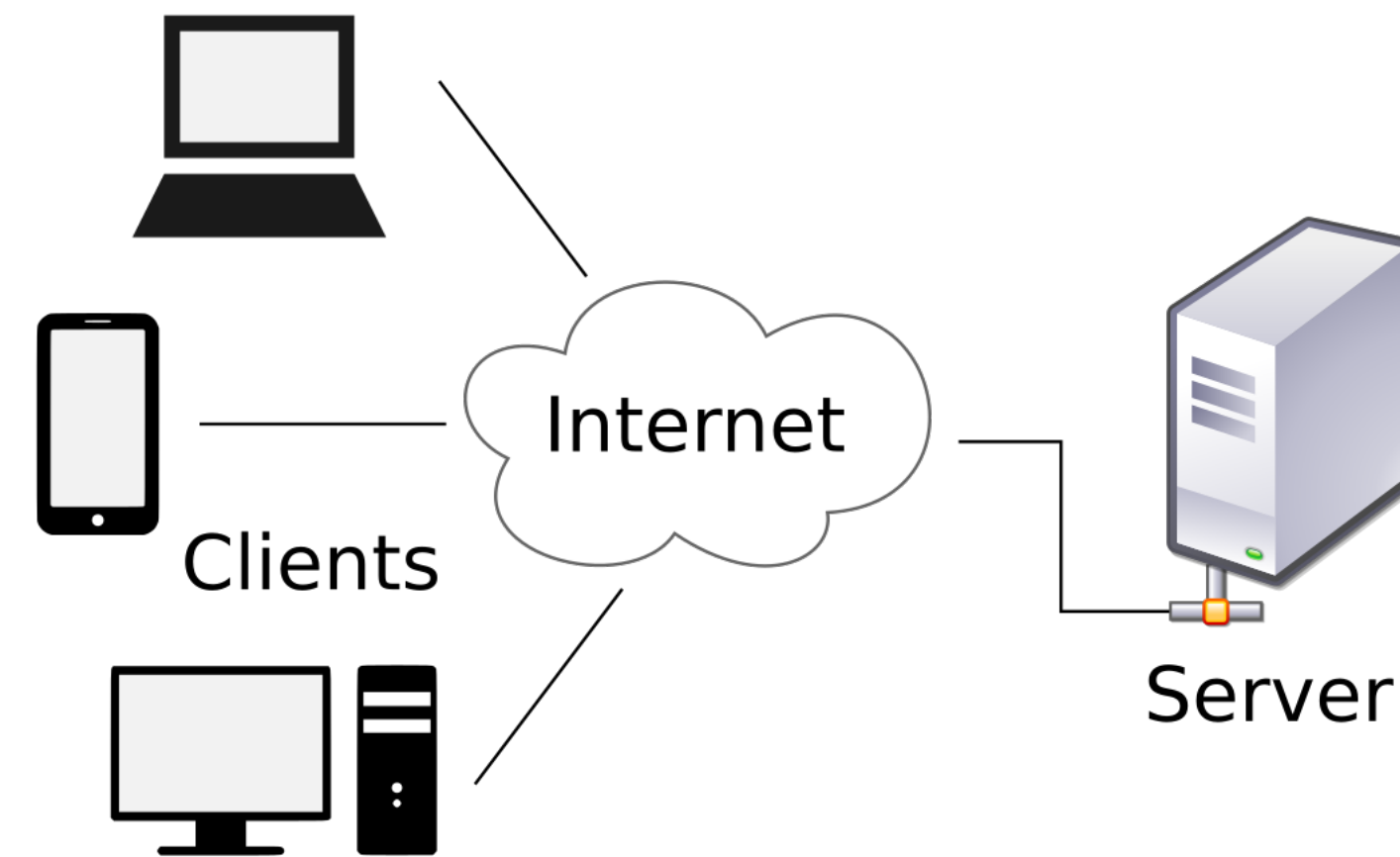
- Client: a user's device (or browser)
- Server: a shared device that manages resources
- Implementing large, complex applications requires both
- But you can often use the same programming languages, tools, etc. for both



So what does any of this have to do with frontends and backends?

Clients, Servers, Frontend, and Backend

- The distinction between clients and servers largely parallels *frontend* versus *backend* development
- Frontend: everything a user sees and interacts with
- Backend: all of the logic that powers the functionality



Frontend development

- Programming the graphical/visual functionality
 - Interactive page elements (drop-downs, menus, etc.)
- Easier (relatively speaking) to start developing for
 - You write a bit of code and it opens up in your browser
- HTML/CSS/JavaScript are pretty standard, but not the only



Frontend development

- Very direct integration with design
 - You will be working most closely with these frontend developers
- Performance matters, but largely in its impact on user experience
 - Do animations look nice? Do pages take too long to load?



Backend development

- Programming the logic and structure; typically nothing visual
 - Databases, authentication (more on this later in the quarter)
- Privacy and security measures are important
 - Enables access to all data, so subject to breaches etc.
- *Concurrency* is really important
 - Multiple frontends are likely connecting to the same backend, often simultaneously
 - What goes first?



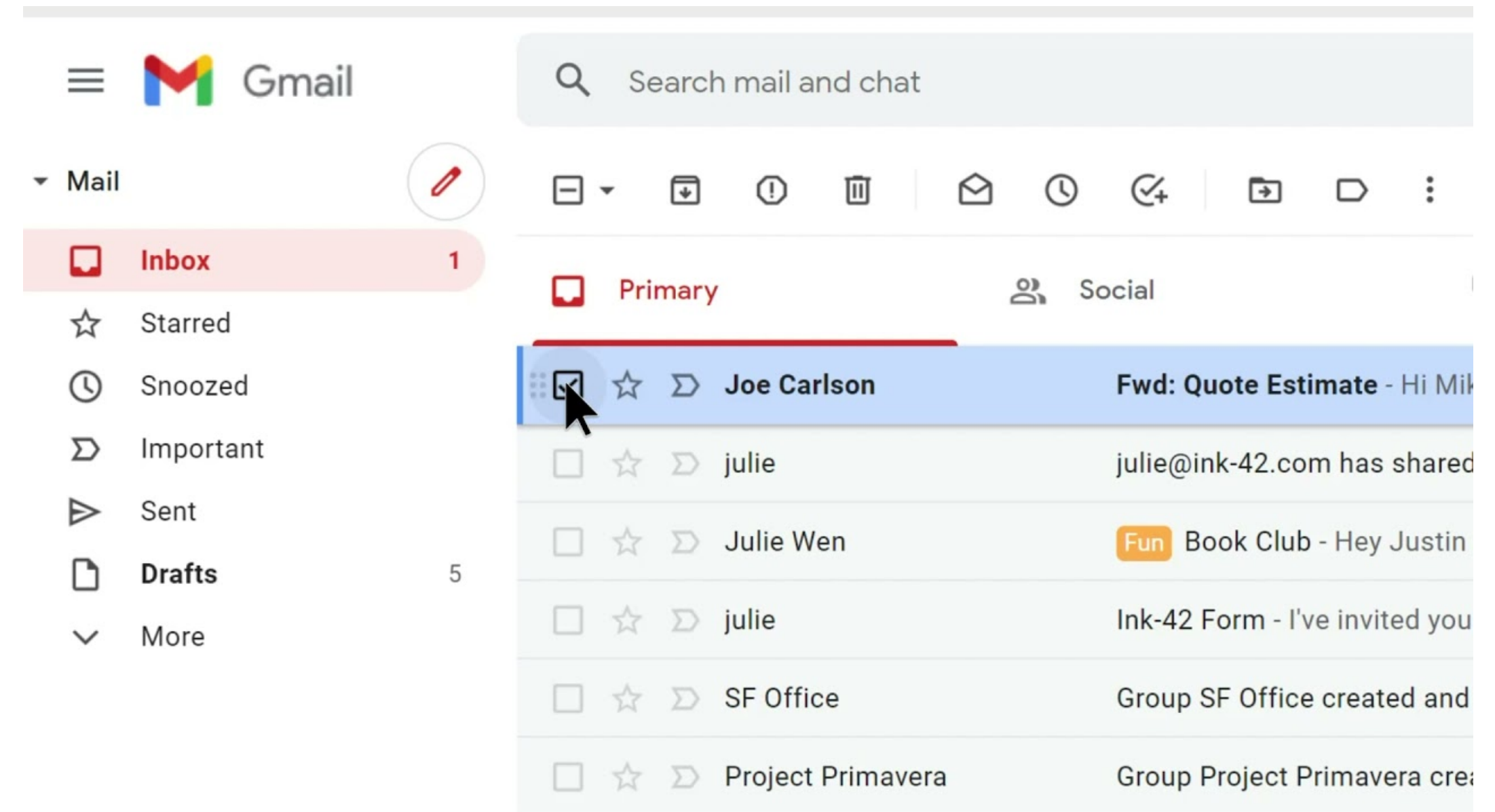
Backend development

- Typically harder to set up
 - Can be made with lots of different programming languages and configurations
- Typically more powerful computers, capable of a lot of computation
 - Speed and efficiency are really important: handling lots of concurrent operations, with a lot of emphasis on optimization and math



One example: Gmail

- Frontend
 - Compose, navigate between inboxes, display of labels/stars/etc.
- Backend
 - Storage of email and drafts, syncing across users, authentication



Frontend or Backend?

- You record a video on your phone. Where are the controls for *editing* that video?



Or...



Editing is done via interfaces, so it will be implemented in the **frontend**.

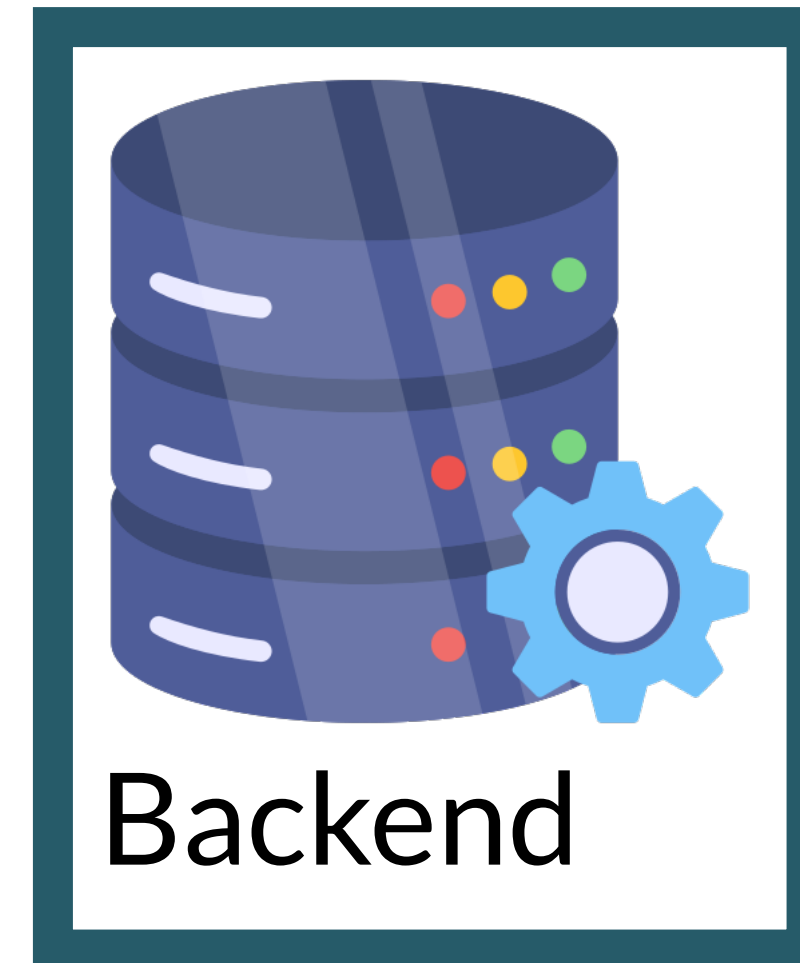
Frontend or Backend?

- Your video has been edited, and is ready to be shared online. What will *host it* for others to see?



Frontend

Or...



Backend

Backends are typically in charge of storage, especially for ensuring access across people and devices.

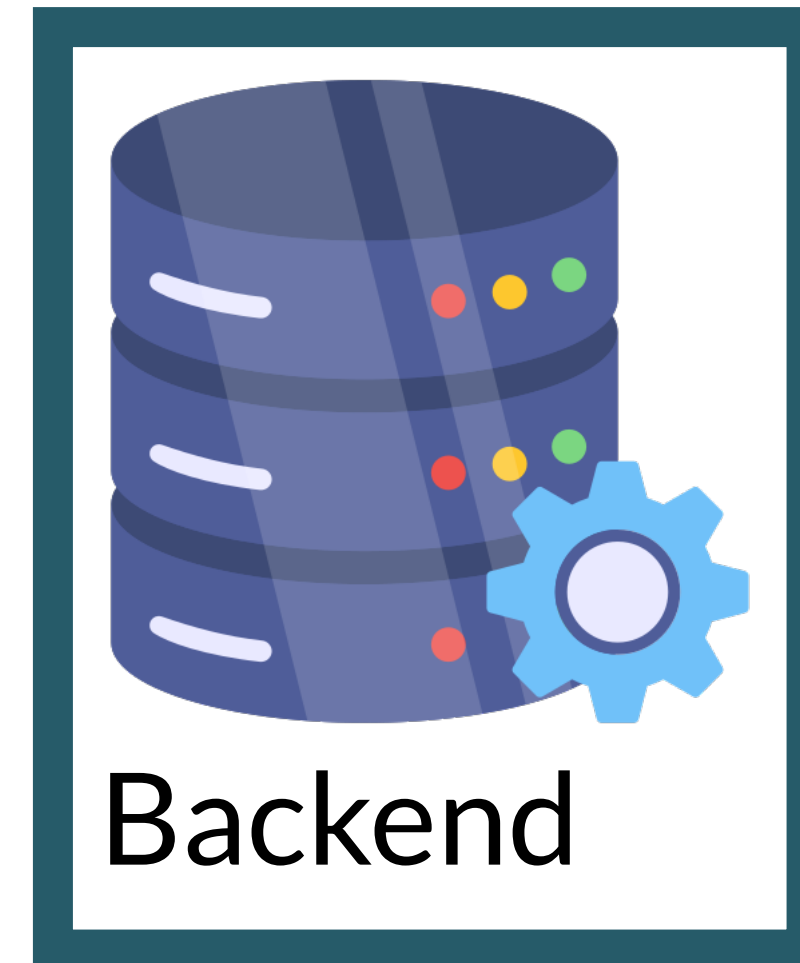
Frontend or Backend?

- You want to add AI-generated captions to your video. Which will *produce* the captions?



Frontend

Or...



Backend

It is computationally expensive to generate AI captions, so a **backend** will probably do it.

Frontend or Backend?

- Your friend uses a screen reader to access your video. What is their reader *processing* to make the content accessible?



Or...



User-setup extensions like screen readers only have access to information visible in the **frontend**.

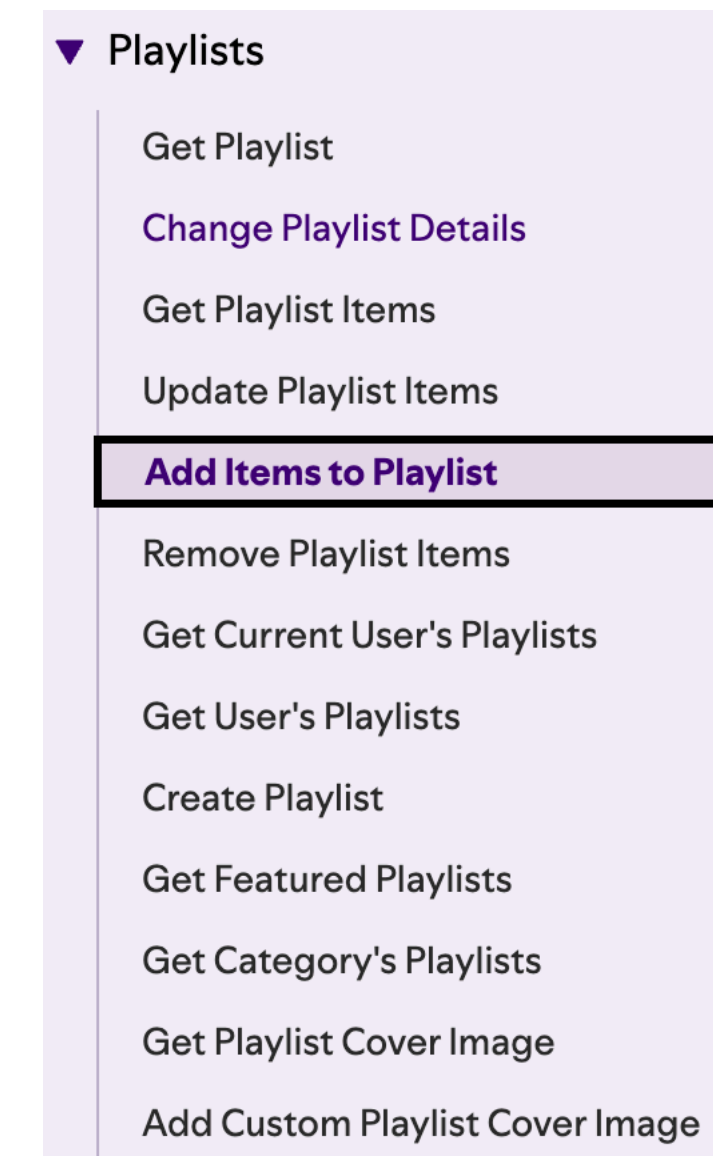
Frontend and Backend Communication

- Frontends and backends are *constantly* communicating with one another, especially in more interactive pages
- HTTP, the same protocol for getting pages from web servers, is used for most communication
- Clear specification is very important

Frontend and Backend Communication

One example: Spotify's API

- API: Application Programming Interface
- Lists what functionality is available in the backend
 - Provides specification for exactly how to tell the backend what to do
- In your frontend, you can then follow that specification



Web API • References / Playlists / Add Items to Playlist

Add Items to Playlist OAuth 2.0

Add one or more items to a user's playlist.

Authorization scopes

- ▶ playlist-modify-public
- ▶ playlist-modify-private

Request

▼ **POST** **/playlists/{playlist_id}/tracks**

playlist_id string Required

The [Spotify ID](#) of the playlist.

Example: `3cEYpjA9oz9GiPac4AsH4n`

Frontend and Backend in practice

- Most applications need both a frontend and a backend
- In small projects, a developer might do both (“full stack” development)
- Larger projects typically divide the roles, and developers specialize

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